Q-ORACLE: Semantic-Embedding Temporal Quantum Machine Learning for PESTEL Scenario Forecasting

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Abstract

We present Q-ORACLE, a prototype pipeline that extends weekly news-graph foresight into temporal quantum machine learning (QML) experiments on real hardware. The system starts from ORACLE-style time-dependent recursive summary graph snapshots, extracts weekly PESTEL world-state vectors, builds semantic embeddings from clustered weekly evidence, projects the embeddings into hardware-sized quantum feature maps, trains shallow variational circuits on week-to-week transitions, and submits the learned inference circuits to IQM Sirius hardware. The updated experiment uses three weekly graph snapshots, `intfloat/multilingual-e5-large-instruct` 1024-dimensional semantic embeddings, 8-, 10-, and 12-qubit latent circuits, two variational layers, and a two-bit scenario readout. We executed 18 IQM Sirius semantic-QML jobs across three seeds, two shot counts and three circuit sizes with no failed submissions. The aggregate hardware distribution was 31.5% stable, 29.5% event-aligned, 19.7% event-divergent and 19.4% uncertain. A compact four-qubit PESTEL-only circuit is retained as a baseline. We position the result as a reproducible hardware-backed systems demonstration rather than evidence of predictive market validity.

1 Introduction

News-driven foresight systems are useful only when they preserve traceability from raw evidence to the resulting strategic interpretation. ORACLE-style time-dependent recursive summary graphs (TRSGs) address this by converting news streams into weekly clustered graph snapshots, summarizing them recursively, and grounding interpretation in PESTEL dimensions. The reference ORACLE paper frames this as an auditable foresight layer rather than an opaque prediction engine [1].

Q-ORACLE asks a narrower experimental question: can the weekly PESTEL trajectory produced by such a foresight system be converted into a small temporal QML task and executed on real quantum hardware? The goal is not to claim financial prediction or quantum advantage. Instead, the goal is to make the whole chain concrete enough for inspection: graph snapshots become vectors, vectors become training pairs, training pairs define a variational circuit, and the trained inference circuit is sampled on a quantum processing unit (QPU).

The resulting prototype runs as a standalone backend/frontend system with a hardware folder for provider-specific execution. It now contains the original compact PESTEL hardware receipts plus an 18-job semantic-embedding QML batch on IQM Sirius. This paper documents the method, implementation, hardware results and limitations.

2 Related Work

News foresight and PESTEL. ORACLE builds a weekly TRSG from crawled news, embeddings, hierarchical clustering, recursive summarization and PESTEL-aware change detection [1]. PESTEL analysis has also been explored with large language models for forecasting changes in operating environments [2]. Our work keeps the PESTEL framing but replaces the final interpretation layer with a hardware-executed temporal QML experiment.

Quantum time-series learning. Variational quantum circuits have been proposed for one-step time-series forecasting [3], multidimensional generative time-series modeling [4], and long-term multivariate forecasting [5]. Recent benchmarks caution that variational quantum methods often struggle to outperform simple classical baselines under fair comparison [6]. We therefore treat Q-ORACLE as an empirical pipeline demonstration, not as an advantage claim.

Feature maps and trainability. Quantum machine learning can be interpreted as learning in quantum feature spaces [7]. However, trainability of variational circuits remains an open challenge; barren plateaus and noise can flatten optimization landscapes [8]. Q-ORACLE therefore tests shallow circuits and explicit dimensionality reduction: a compact four-qubit PESTEL baseline and semantic-embedding circuits with 8, 10 and 12 latent qubits.

3 System Overview

Figure 1 summarizes the pipeline. Weekly graph snapshots are normalized into PESTEL vectors and also converted into semantic embedding vectors from clustered weekly evidence. A temporal forecaster produces scenario branches. Event text is converted into an interest vector. The compact QML module compresses PESTEL and event information into four latent variables. The semantic QML module projects 1024-dimensional weekly embeddings into 8-, 10- and 12-qubit feature maps, trains shallow variational circuits on week transitions and submits the resulting inference circuits to IQM.

3.1 Input Data

The current artifact uses three weekly snapshots stored in the QuantumFinances hardware payload. Each week contains graph metadata, clusters, raw graph links and a normalized PESTEL vector:

$$v_t = [P_t, E_t, S_t, T_t, Env_t, L_t] \in [0,1]^6.$$

The latest run payload also contains an event vector and scenario probabilities produced by the Q-ORACLE scenario engine. For the semantic experiment, clustered weekly text and the event query are embedded with `intfloat/multilingual-e5-large-instruct`. The resulting embedding dimension is 1024, as recorded in the generated embedding payload.

3.2 PESTEL Timeline

Figure 2 shows the weekly trajectory used by the experiment. The values are not raw market data; they are normalized world-state features derived from clustered news snapshots. The latest vector is listed in Table 2.

4 Compact PESTEL QML Baseline

4.1 Latent Compression

To keep the QPU circuit small, the six-dimensional PESTEL vector is compressed into four latent features:

$$z_t = [z_t^{PL}, z_t^{ET}, z_t^{SE}, z_t^{G}].$$

The first three combine semantically paired PESTEL dimensions: political/legal, economic/technological and social/environmental. The fourth combines temporal trend and graph intensity:

$$z_t^G = \mathrm{clip}(0.50 + 0.75\bar{\Delta}_t + 0.35 g_t),$$

where $\bar{\Delta}_t$ is the mean week-over-week PESTEL change and g_t is graph link intensity.

4.2 Training Objective

For each transition $t \to t+1$, the model builds a four-class target distribution:

$$y_t = [p_{\text{stable}}, p_{\text{up}}, p_{\text{down}}, p_{\text{uncertain}}].$$

The target uses event-vector alignment, weighted PESTEL similarity and volatility. The circuit output distribution $q_\theta(z_t)$ is optimized using cross-entropy:

$$\mathcal{L}(\theta) = -\sum_t \sum_c y_{t,c} \log q_{\theta,c}(z_t).$$

Because the available history is short, the optimizer is deliberately simple: seeded local stochastic search over a compact 16-parameter ansatz. This makes the run reproducible and avoids overpresenting the result as a trained forecasting model.

4.3 Circuit

The circuit has four data qubits and two measured scenario bits. Each latent feature is encoded through R_y and R_z rotations. Two variational layers then apply per-qubit rotations and a ring of CNOT entanglers. The final two measured qubits define scenario classes:

$$\begin{aligned} &00 = \text{stable}, && 01 = \text{aligned up}, \\ &10 = \text{aligned down}, && 11 = \text{uncertain}. \end{aligned}$$

5 Semantic Embedding QML Method

5.1 Embedding Construction

The semantic experiment uses the same weekly graph snapshots but treats cluster text as the primary representation. For each week, cluster texts are embedded with `intfloat/multilingual-e5-large-instruct`. Cluster embeddings are L2-normalized and averaged using cluster size weights to create one weekly vector:

$$e_t \in \mathbb{R}^{1024}.$$

The event query is embedded with the same model, yielding $e_{\text{event}} \in \mathbb{R}^{1024}$.

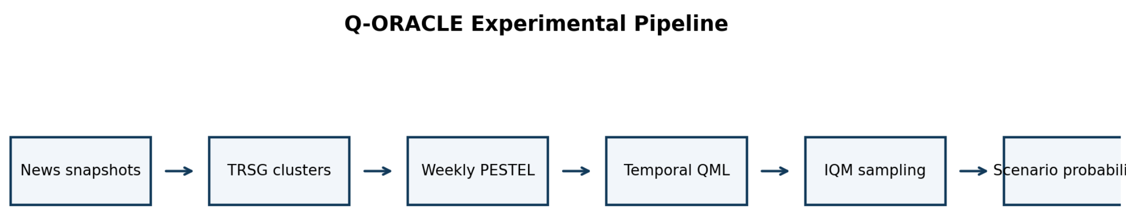


Figure 1: Q-ORACLE experimental pipeline from news snapshots to quantum-sampled scenario probabilities.

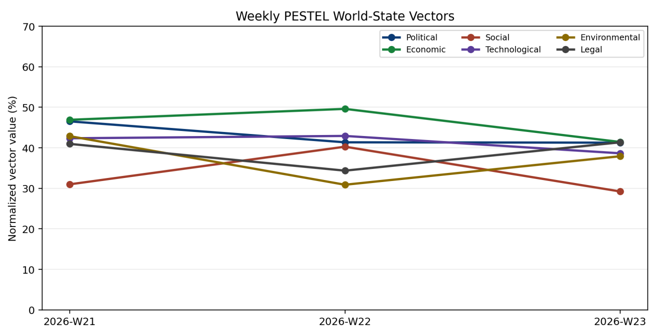


Figure 2: Weekly PESTEL vectors for the current sample payload.

5.2 Projection Into Quantum Features

The QPU does not receive 1024 physical qubits. Instead, the semantic vectors are projected into hardware-sized latent feature vectors:

$$x_t^{(d)} = \Pi_d(e_t), \qquad d \in \{8, 10, 12\}.$$

The implemented projection uses PCA fitted on the weekly/event embedding set and seeded signed-random components when the requested latent dimension is larger than the available PCA rank. Values are scaled through a bounded nonlinearity into $[0, 1]$, then encoded as R_y and R_z rotations.

5.3 Semantic Transition Target

For each transition $t \to t + 1$, the target distribution is derived from semantic similarity to the event embedding and week-to-week drift:

$$y_t = [p_{\text{stable}}, p_{\text{event}}, p_{\text{divergent}}, p_{\text{uncertain}}].$$

Here event-aligned means the next weekly embedding moves closer to the event embedding, event-divergent means it moves away, and uncertain is driven by embedding drift. The variational circuit uses two layers of rotations and ring entanglement and measures two scenario bits.

6 Implementation

The implementation is contained in the QuantumFinances repository. The compact temporal QML hardware script is:

`quantum_hardware/scripts/iqm_temporal_qml_submit.py`

It consumes `latest_run.json`, trains the variational parameters, writes a receipt and optionally submits to IQM Resonance through Qiskit-on-IQM. A batch runner, `iqm_batch_qml_experiments.py`, repeats the experiment over seeds and shot counts, writing JSON and CSV summaries.

The semantic-embedding path is implemented by:

`build_semantic_embeddings.py`

and

`iqm_semantic_embedding_qml_submit.py`.

The semantic batch runner is:

`iqm_batch_semantic_qml_experiments.py`

It repeats the experiment over seeds, shot counts and latent circuit sizes.

The run command used for the verified compact temporal-QML experiment was:

```
iqm_temporal_qml_submit.py --input latest_run.json --shots 128
--iterations 60 --submit --wait
```

7 Hardware Results

The compact temporal QML job was submitted to IQM Sirius and returned job ID:

`019e9de9-956e-7622-ace6-80049a84a06c`.

The result contains 128 shots with counts shown in Table 3. The local statevector distribution and hardware sample distribution are compared in Table 1 and Figure 3.

The semantic-embedding QML batch was then executed on IQM Sirius using 1024-dimensional E5 embeddings projected into 8-, 10- and 12-qubit latent circuits. The batch contains 18 submitted hardware jobs and 13,824 total hardware shots. Table 6 summarizes the configuration, Table 7 reports mean hardware distributions by circuit size, and Table 8 reports the aggregate distribution across all semantic jobs.

Class	Local QML	IQM hardware
Stable	49.4%	41.4%
Aligned Up	26.2%	20.3%
Aligned Down	9.4%	25.0%
Uncertain	15.0%	13.3%

Table 1: Temporal QML scenario distribution. The local column is the statevector prediction after classical training; the hardware column is sampled from IQM Sirius.

Dimension	Latest value
Political	41.3%
Economic	41.4%
Social	29.2%
Technological	38.6%
Environmental	37.9%
Legal	41.4%

Table 2: Latest weekly PESTEL vector used for feature encoding.

Bitstring	Count	Share
00	53	41.4%
01	26	20.3%
10	32	25.0%
11	17	13.3%

Table 3: Raw IQM temporal-QML hardware counts.

Experiment	Job ID	Shots
Feature circuit	019e9de2-2ab7-7653-9134-15e34fbe5926	128
Temporal QML	019e9de9-956e-7622-ace6-80049a84a06c	128

Table 4: Verified IQM hardware jobs available at paper-generation time.

Batch mode	Dry-run
Records	10
Total shots	960

Table 5: Automatic temporal-QML batch summary used for the seed-sensitivity figure.

Semantic embedding model	`intfloat/multilingual-e5-large-instruct`
Embedding dimension	1024
Hardware jobs	18
Total hardware shots	13824
Latent qubits tested	8, 10, 12

Table 6: Semantic-embedding QML batch configuration.

Qubits	Records	Shots	Stable	Event aligned	Event divergent	Uncertain
8	6	4608	32.3%	29.3%	19.1%	19.2%
10	6	4608	29.3%	27.6%	21.8%	21.2%
12	6	4608	32.7%	31.4%	18.0%	17.9%

Table 7: Mean IQM Sirius hardware distributions for the 1024-dimensional semantic-embedding QML batch.

Class	Overall mean
Stable	31.5%
Event Aligned	29.4%
Event Divergent	19.7%
Uncertain	19.4%

Table 8: Aggregate distribution across all semantic-embedding QML hardware jobs.

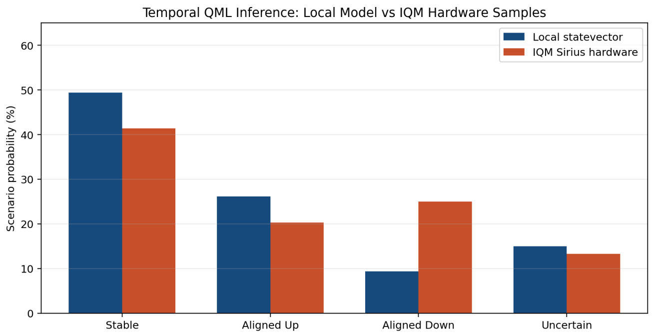


Figure 3: Temporal QML local inference probabilities compared with IQM Sirius hardware samples.

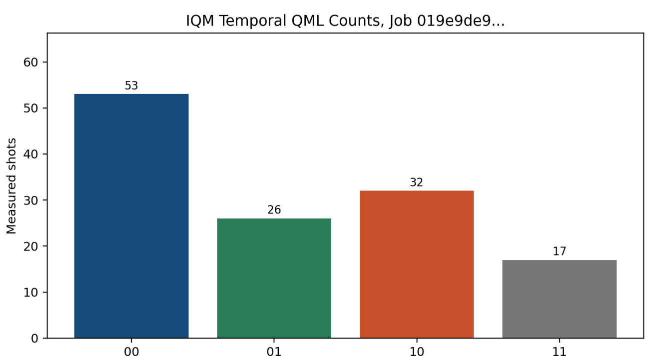


Figure 4: Raw IQM temporal QML counts for the verified hardware job.

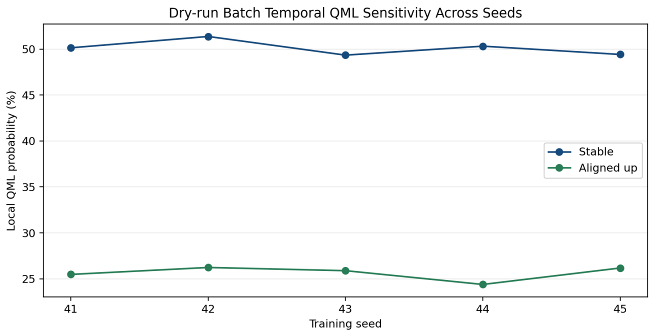


Figure 5: Compact temporal-QML baseline sensitivity across seeds.

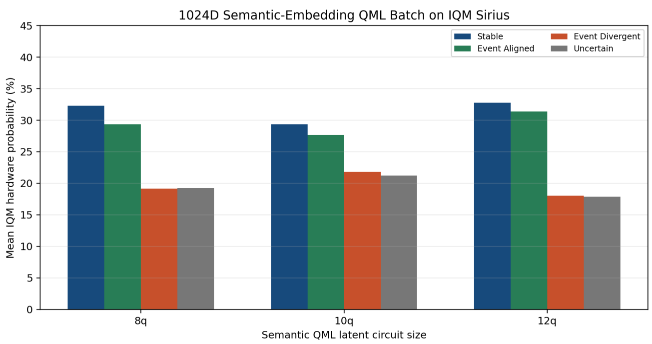


Figure 6: Mean IQM Sirius distributions for 1024-dimensional semantic-embedding QML projected into 8-, 10- and 12-qubit circuits.

8 Batch Experiments

The compact PESTEL batch runner provides a local seed-sensitivity baseline. Figure 5 shows this baseline result. The semantic batch is the main updated experiment: three seeds, two shot counts and three latent circuit sizes, all submitted to IQM Sirius. Figure 6 shows mean hardware probabilities by circuit size, while Figure 7 shows the event-aligned signal across shot counts.

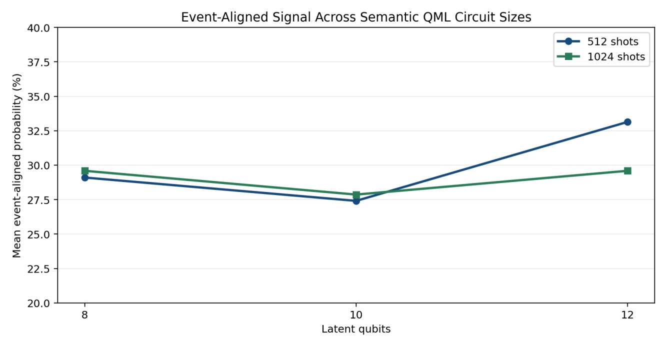
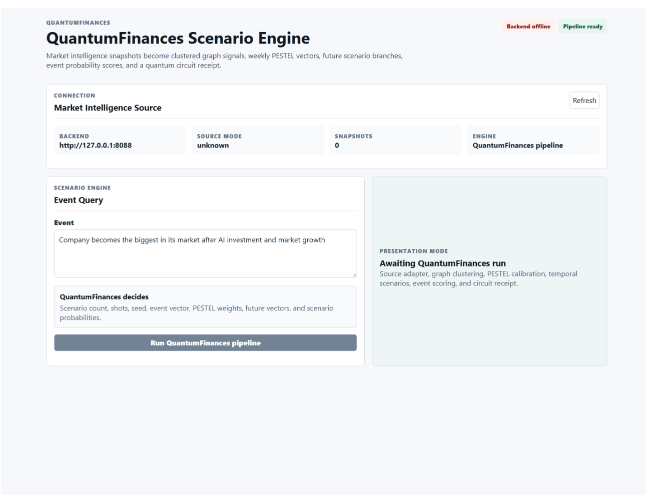


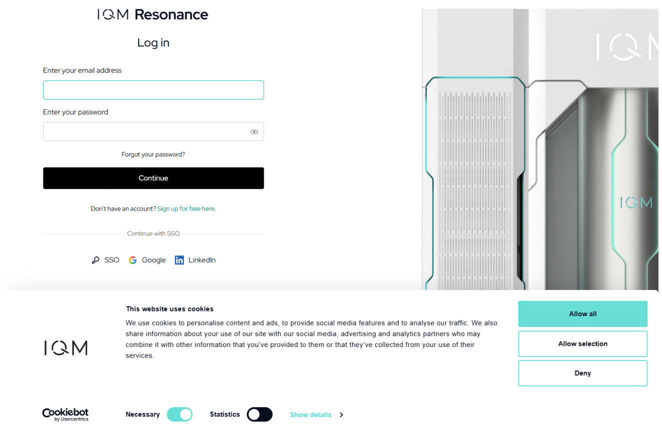
Figure 7: Event-aligned semantic-QML probability by latent circuit size and shot count.

9 Screenshots and Reproducibility Artifacts

Figure 8 captures the working prototype interface and IQM Resonance access point used for hardware execution. These screenshots are not used as scientific evidence; they document the operational environment for the hackathon artifact.



(a) Q-ORACLE scenario interface.



(b) IQM Resonance public access page.

Figure 8: Screenshots included for reproducibility and presentation context.

10 Discussion

The hardware runs are informative in three ways. First, they verify that the ORACLE/PESTEL pipeline can produce circuits that survive provider transpilation and execute on a real superconducting QPU. Second, the semantic batch demonstrates that high-dimensional text embeddings can be used as the data source while the QPU receives only an explicit, auditable latent projection. Third, the repeated hardware jobs expose seed, shot and circuit-size sensitivity instead of hiding uncertainty behind a single simulated receipt.

Across all 18 semantic hardware jobs, stable remains the largest aggregate branch at 31.5%, but event-aligned is close at 29.5%. The 12-qubit setting produced the strongest event-aligned mean among the tested circuit sizes. This should be interpreted as a hardware-sampled scenario signal under the current embedding/projection design, not as a calibrated probability of a real-world event.

11 Limitations

The most important limitation is data scale: three weekly snapshots provide only two transition examples. This is enough for a hackathon hardware demonstration but not enough for statistical validation. The semantic embeddings are real 1024-dimensional model outputs, but they are computed from a small sample payload. The projection from 1024 dimensions to 8–12 qubits is explicit and reproducible, yet it is still an information bottleneck. Additionally, the circuits are shallow by design; larger ansatzes may be more expressive but are harder to train and noisier on near-term hardware. Finally, the event probability is a scenario compatibility score, not investment advice or a verified forecast.

12 Conclusion

Q-ORACLE demonstrates an end-to-end path from auditable news graph foresight to temporal QML execution on real IQM hardware. The updated prototype turns weekly PESTEL vectors and 1024-dimensional semantic embeddings into transition targets, trains compact variational circuits and samples scenario registers on IQM Sirius. The main contribution is not predictive accuracy, but a reproducible architecture for linking narrative intelligence, semantic vector modeling and quantum hardware receipts.

References

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